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Course ID: 1142-1764

Foundations of Computer Science

02-Data and Database

Course Info

Introduction to Computer Science

Fundamentals · Organization · Networks · OS · Algorithms · Data Structures · ADT

2 Semesters · 11 Topics · 32 Weeks · 3 Credits each

INSTRUCTOR	DEPARTMENT	LANGUAGE	TOOLS
Prof. H.T. Chu	**International Program in Computer Science**	**English (EMI)**	**AWS Academy**

ASSESSMENT (EACH SEMESTER)

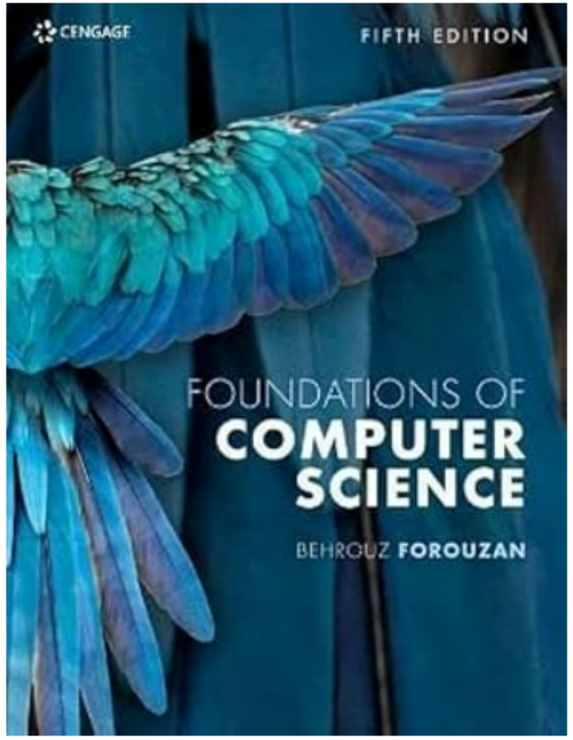
Assignments	Midterm Exam	Project / Labs	Final Exam	Participation
30%	**20%**	**30%**	**20%**	**Bonus 5%**

Providence University · Department of Computer Science and Information Engineering · AY 2025–2026

Thursday, 3-4 (10:10-12:00)

Providence Hall R307

The Textbook



Foundations of Computer Science Paperback

作者 Behrouz Forouzan (Author)

The symposium was held Oct.-Nov. 1989, Research Triangle Park, North Carolina. One hundred papers in theoretical computer science treat dispersers, deterministic amplification, and weak random sources; efficient NC algorithms for set cover with applications to learning and geometry; the inverse of automorphism in polynomial time; and speeding-up linear programming using fast matrix multiplication. Acidic paper; no subject index. Annotation copyright Book News, Inc. Portland, Or.

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語言	尺寸	ISBN-10	ISBN-13
English	18.9 x 24.7 x 2.6 cm	1473787327	978-1473787322

查看所有詳細內容

1 Introduction

Turing Model
Von Neumann Model
Computer Components
History
Social and Ethical Issues
Computer Science as a Discipline
Outline of the course
End-Chapter Materials
Practice Set

2 Number Systems

Introduction
Positional Number Systems
Non-Positional Number Systems
End-Chapter Materials
Practice Set

3 Data Storage

Data Types
Storing Numbers
Storing Text
Storing Audio
Storing Images
Storing Video
End-Chapter Materials
Practice Set

4 Operations on Data

Logic Operations
Shift Operations
Arithmetic Operations
End-Chapter Materials
Practice Set

5 Computer Organization

Introduction
Central Processing Unit
Main Memory
Input/Output Subsystem
Subsystem Interconnection
Program Execution
Different Architectures
A Simple Computer
End-Chapter Materials
Practice Set

6 Computer Networks

Introduction
TCP/IP Protocol Suite
Layers
Internet Applications
The World Wide Web (www)
End-Chapter Materials
Practice Set

7 Operating Systems

Introduction
Evolution
Components
A Survey of Operating Systems
End-Chapter Materials
Practice Set

8 Algorithms

Concept
Three constructs
Algorithm Representation
A More Formal Definition
Basic Algorithms
Subalgorithms
Recursion
End-Chapter Materials
Practice Set

9 Efficiency of Algorithms

Definition
Common Algorithms
Big-O Notation
End-Chapter Materials
Practice Set

10 Programming Languages

Evolution
Translation
Programming Paradigms
Common Concepts
End-Chapter Materials
Practice Set

11 Software Engineering

The Software Life cycle
Analysis Phase
Design Phase
Implementation Phase
Testing Phase
Documentation
End-Chapter Materials
Practice Set

12 Data Structure

Arrays
Records
Linked Lists
End-Chapter Materials
Practice Set

13 Abstract Data Types

Background
Stacks
Queues
General Linear Lists
Trees
Graphs
End-Chapter Materials
Practice Set

This semester

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Graphs

+ aws academy

1

HOUR

10:10 – 11:00

Topic

Data & Databases

Data & Databases

Welcome to the course — let's get started.

Week 2

50 minutes

Number Systems

Binary & Hexadecimal

How computers speak — and how we read them

0110 1000 0101 1100 1010 0x1F 0xA3 0xFF

Computer Science Fundamentals

01 *Definitions, types, and examples*

02 *Bits, bytes, and encoding*

03 *The DIKW pyramid*

04 *Organizing information*

05 *Why DBMS matters*

06 *Tables, queries, and joins*

07 *Modern data challenges*

What Is Data?

"Data is a collection of raw, unprocessed facts, symbols, or observations that represent real-world values."

Numbers	Text	Boolean	Date/Time
42, 3.14, -7	*"Hello", "Alice"*	*True / False*	*2025-03-05 14:30*

Data alone has no meaning — it must be processed and interpreted.

Common Data Types

Integer

`int: 0, 42, -100`

Whole numbers, positive or negative

Float

`float: 3.14, -0.001`

Numbers with decimal points

String

`str: "hello", "CS101"`

Sequence of characters / text

Boolean

`bool: True, False`

Logical values — only two states

List/Array

`[1, 2, 3], ['a','b']`

Ordered collection of values

Dictionary

`{"name":"Alice","age":20}`

Key-value pairs of data

All data inside a computer is stored as binary — a series of 0s and 1s.

1	0	1	1	0	0	1	0

← 1 Byte = 8 Bits → (this byte = 0xB2 = 178 in decimal)

Unit	Size	Example
1 Bit	0 or 1	On/Off switch
1 Byte	8 bits	1 character 'A'
1 Kilobyte (KB)	1,024 bytes	Short text file
1 Megabyte (MB)	1,024 KB	~1 minute of music
1 Gigabyte (GB)	1,024 MB	~700 photos
1 Terabyte (TB)	1,024 GB	~250,000 songs

Character Encoding: ASCII & Unicode

Computers don't natively store letters — they store numbers that map to characters.

Char	Dec	Bin
A	65	01000001
B	66	01000010
C	67	01000011
a	97	01100001
0	48	00110000
!	33	00100001

ASCII

American Standard Code for Information Interchange. 128 characters, 7-bit encoding. Covers English letters, digits, punctuation.

Unicode / UTF-8

Universal standard supporting 143,000+ characters from all world languages, emoji, and special symbols.

From Data to Knowledge: The DIKW Pyramid

WISDOM
Knowing why — judgement & insight

KNOWLEDGE
Understanding patterns & meaning

INFORMATION
Data with context and structure

DATA
Raw facts & unprocessed observations

Increas
ng
Value &
Contex

Structured vs. Unstructured Data

Structured Data	Unstructured Data
Organized in rows & columns	No fixed format or schema
Follows a defined schema	Varied formats & types
Stored in relational databases	Stored in files, NoSQL, or data lakes
Easy to search & analyze	Harder to search directly
Examples: bank records, CSV files, spreadsheets	Examples: emails, videos, social media posts, images

💡 Semi-Structured Data (e.g. JSON, XML) has flexible schema — a middle ground between structured and unstructured.

Introduction to Databases

A database is an organized collection of structured data stored electronically, designed for efficient access, management, and updating.

Organized Storage

Data is stored in a systematic, logical structure

Fast Retrieval

Indexes allow rapid searching through millions of records

Data Integrity

Rules and constraints ensure accuracy and consistency

Multi-User Access

Many users can access data simultaneously

Data Independence

Applications can change without restructuring data

Backup & Recovery

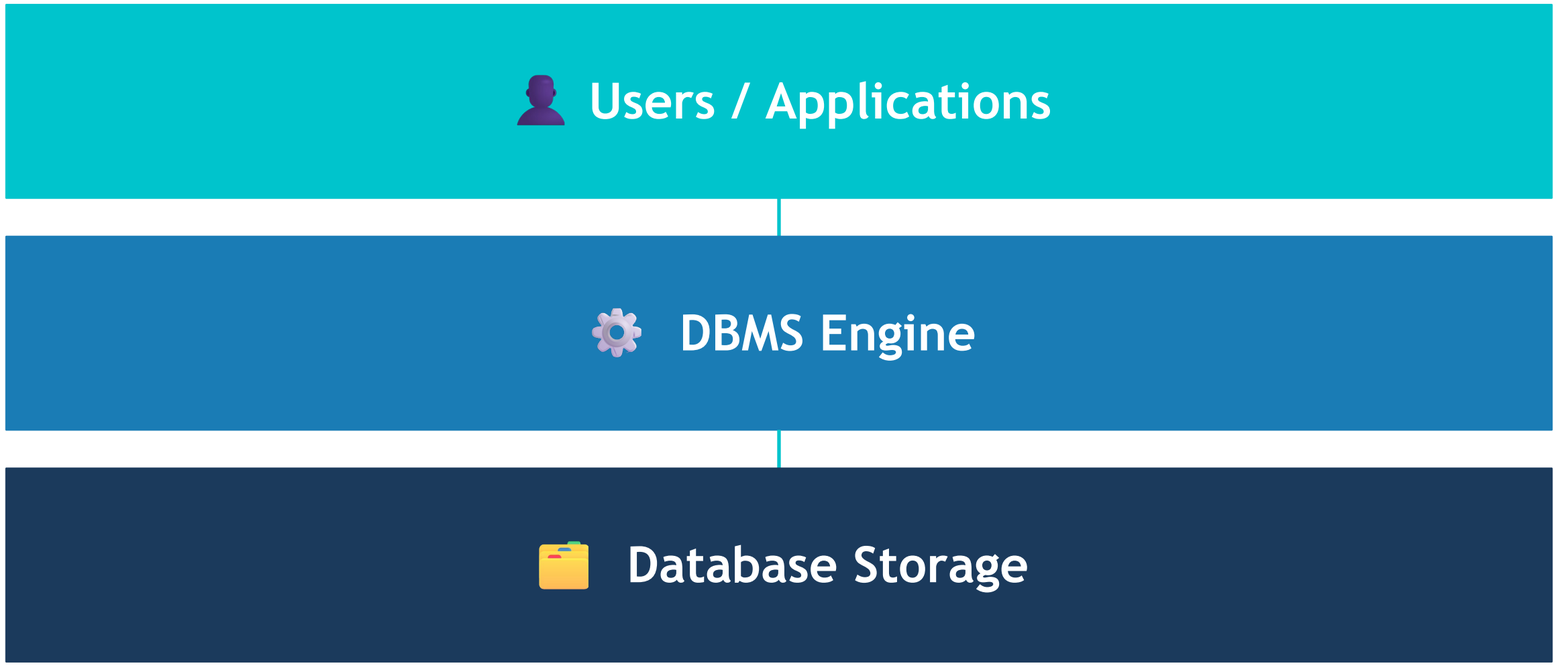
Protection against loss through transactions and logs

File Systems vs. Database Systems

Feature	File System	Database System
Data Redundancy	High — same data duplicated	Controlled & minimized
Data Integrity	No automatic enforcement	Built-in constraints & rules
Concurrency	Limited / problematic	Full concurrent access support
Security	OS-level file permissions only	Fine-grained access control
Querying	Manual code required	Declarative SQL queries
Recovery	Manual backup only	Automatic transaction logs

Database Management Systems (DBMS)

A DBMS is software that acts as an intermediary between users and the database.



Popular DBMS Examples

MySQL	PostgreSQL	Oracle DB	MongoDB	SQLite
Relational	Relational	Relational	NoSQL	Embedded

Relational Databases

Data is organized into tables (relations) with rows and columns, connected through shared keys.

Table: Students

StudentID	Name	Age	Major
S001	Alice	20	CS
S002	Bob	21	Math
S003	Carol	22	CS

Table: Enrollments

EnrollID	StudentID	Course	Grade
E001	S001	CS101	A
E002	S001	CS201	B+
E003	S002	MATH101	A-

Primary Key: Unique identifier for each row (e.g., StudentID)

Foreign Key: Column that links to another table's primary key

Table / Relation: Grid of data with defined rows and columns

Schema: Blueprint defining the structure of the database

Structured Query Language (SQL)

SQL is the standard language for interacting with relational databases.

DDL — Data Definition	DML — Data Manipulation	DQL — Data Query	DCL — Data Control
CREATE TABLE	INSERT INTO	SELECT … FROM	GRANT
ALTER TABLE	UPDATE SET	WHERE clause	REVOKE
DROP TABLE	DELETE FROM	ORDER BY	DENY

```
-- Find all CS students older than 20
SELECT Name, Age FROM Students
WHERE Major = 'CS' AND Age > 20
ORDER BY Name ASC;
```

SQL CRUD Operations

CRUD = Create · Read · Update · Delete — the four basic operations on data

CREATE — Add new records to a table

```
INSERT INTO Students (StudentID, Name, Age)
VALUES ("S004", "David", 19);
```

READ — Retrieve and display existing data

```
SELECT * FROM Students
WHERE Major = "CS";
```

UPDATE — Modify existing records

```
UPDATE Students
SET Age = 21
WHERE StudentID = "S001";
```

DELETE — Remove records from a table

```
DELETE FROM Students
WHERE StudentID = "S003";
```

Keys & Relationships in Databases

🔑

Primary Key (PK)

Uniquely identifies each record in a table. Cannot be NULL or duplicate.

e.g. StudentID in Students table

🔗

Foreign Key (FK)

A field in one table that references the primary key of another table.

e.g. StudentID in Enrollments table

🔐

Composite Key

A primary key made of two or more columns combined to ensure uniqueness.

e.g. (StudentID + CourseID) together

✅

Unique Key

Ensures all values in a column are distinct, but can allow one NULL value.

e.g. Email address in Users table

Database Normalization

Normalization organizes data to reduce redundancy and improve data integrity.

1NF

First Normal Form:

All columns contain atomic (indivisible) values. No repeating groups.

2NF

Second Normal Form:

Must be in 1NF. Every non-key column depends on the entire primary key.

3NF

Third Normal Form:

Must be in 2NF. No transitive dependencies between non-key columns.

BCNF

Boyce-Codd NF:

Stronger version of 3NF. Every determinant must be a candidate key.

🎯 **Goal: Each piece of data should be stored once. Eliminate data anomalies.**

NoSQL Databases

NoSQL = 'Not Only SQL' — flexible, scalable databases for modern, large-scale applications.

📄 Document Store

Stores data as JSON-like documents. Flexible schema.

Ex: MongoDB, CouchDB

Use: Catalogs, content management

🔑 Key-Value Store

Simple key-to-value mapping. Extremely fast lookups.

Ex: Redis, DynamoDB

Use: Sessions, caching, real-time

📊 Column-Family

Stores data in column families instead of rows.

Ex: Apache Cassandra, HBase

Use: Big data, analytics, IoT

🕸 Graph Database

Nodes and edges for highly connected data.

Ex: Neo4j, Amazon Neptune

Use: Social networks, fraud detection

Big Data: The 5 V's

Big Data refers to datasets too large and complex for traditional database systems to handle.

📦

Volume

Massive amounts of data — terabytes to exabytes generated daily

⚡

Velocity

Speed at which data is generated, processed, and analyzed in real-time

🎨

Variety

Different types of data: text, images, video, audio, sensor data

✅

Veracity

Uncertainty and quality of data — not all data is reliable or accurate

💎

Value

Turning raw data into actionable insights and business value

📈 **Every day: 2.5 quintillion bytes of data are created worldwide**

Databases in the Real World

🏦 Banking & Finance

Account balances, transactions, fraud detection logs

🏥 Healthcare

Patient records, test results, appointment scheduling

🛒 E-Commerce

Product catalogs, orders, inventory, user reviews

📱 Social Media

User profiles, posts, connections, activity feeds

✈️ Travel & Transport

Bookings, routes, real-time tracking, pricing

🎓 Education

Student records, grades, course enrollment, learning

Summary

Data

Raw facts stored as bits and bytes; typed as numbers, text, booleans, etc.

Representation

All data is binary; ASCII & Unicode encode characters as numbers.

DIKW

Data → Information → Knowledge → Wisdom through context and insight.

Databases

Organized, managed collections — far superior to flat file storage.

Relational DB

Tables linked by primary/foreign keys; queried with SQL.

SQL

SELECT, INSERT, UPDATE, DELETE are the core operations on relational data.

NoSQL

Document, key-value, column, graph stores for flexible/large-scale needs.

Big Data

The 5 V's: Volume, Velocity, Variety, Veracity, Value.

2

HOUR

11:10 – 12:00

Topics

- Computer
- Cloud Computing

Computer & Cloud Computing

Computer

Hardware
CPU, memory, storage, I/O devices — how a computer works

Software
OS, applications, system architecture layers

Networking
Connecting computers: LAN, WAN, protocols

Evolution
From mainframes → PCs → mobile → cloud

Cloud Computing

What is Cloud?
On-demand computing resources delivered over the internet

Service Models
IaaS, PaaS, SaaS — what each layer provides

Deploy Types
Public, Private, Hybrid — overview

Key Providers
AWS, Microsoft Azure, Google Cloud Platform

Computer Basics	Cloud Overview	Live Demo	Q&A
20 min	20 min	10 min	5 min

Cloud Computing

Deployment Models

3 Core Deployment Types

01	Public Cloud
02	Private Cloud
03	Hybrid Cloud
+	NaaS — Network as a Service

01 Public Cloud

Shared infrastructure over the internet

Third-party cloud providers (AWS, Azure, GCP) own and operate computing infrastructure, delivering services to multiple customers over the internet on a shared, pay-as-you-go basis.

Scalability

Instantly scale up/down to match demand — no hardware planning

Cost Model

Pay-per-use OpEx; zero upfront CapEx required

Maintenance

Provider fully manages updates, patches, and hardware

Availability

99.9%+ global SLA with built-in redundancy

USE CASES

- E-commerce & web apps
- SaaS platforms
- Dev / Test environments
- Big data & analytics
- Content delivery (CDN)

Major Providers: AWS · Microsoft Azure · Google Cloud Platform · IBM Cloud · Oracle Cloud

02 Private Cloud

Dedicated infrastructure for one organization

Infrastructure exclusively provisioned and operated for a single organization — on-premises or hosted by a third party — providing full control, enhanced security, and regulatory compliance.

Control

- Full admin access
- Custom hardware configs
- Tailored network topology
- Proprietary software stack

Security

- Isolated network perimeter
- GDPR / HIPAA compliance
- Data sovereignty
- Air-gapped environments

Performance

- Predictable low latency
- No noisy neighbors
- Dedicated resource pools
- Custom SLA terms

Common sectors: Banking · Healthcare · Government · Defense · Research Institutions

03 Hybrid Cloud

Best of private + public combined

A hybrid cloud integrates private and public cloud environments through secure connectivity, allowing workloads to move between them based on cost, compliance, and performance needs.

PRIVATE CLOUD

- Sensitive data
- Core systems
- Compliance workloads
- Low-latency operations

🔗 *VPN · Direct Connect · API Gateway*

PUBLIC CLOUD

- Burst & peak capacity
- Dev & test workloads
- Big data processing
- Disaster recovery

Key benefits: Agility + Control · Cost optimization · Workload portability · Gradual migration path

NaaS *Network as a Service*

NaaS delivers network infrastructure — bandwidth, routing, firewalls, VPNs, and WAN connectivity — as a cloud-based subscription service. Organizations consume networking capabilities without owning or managing physical equipment.

On-Demand Bandwidth

Scale network capacity instantly — no hardware provisioning cycles

Virtual Firewall & SD-WAN

Deploy security policies and WAN routing entirely via software

Pay-as-you-Use

OpEx subscription model — pay only for what you consume

Centralized Management

Manage your entire network fabric from a unified dashboard

Providers: Cisco Meraki · VMware VeloCloud · Aryaka · Zscaler · Cato Networks · AT&T NetBond

Summary

Comparing all deployment models

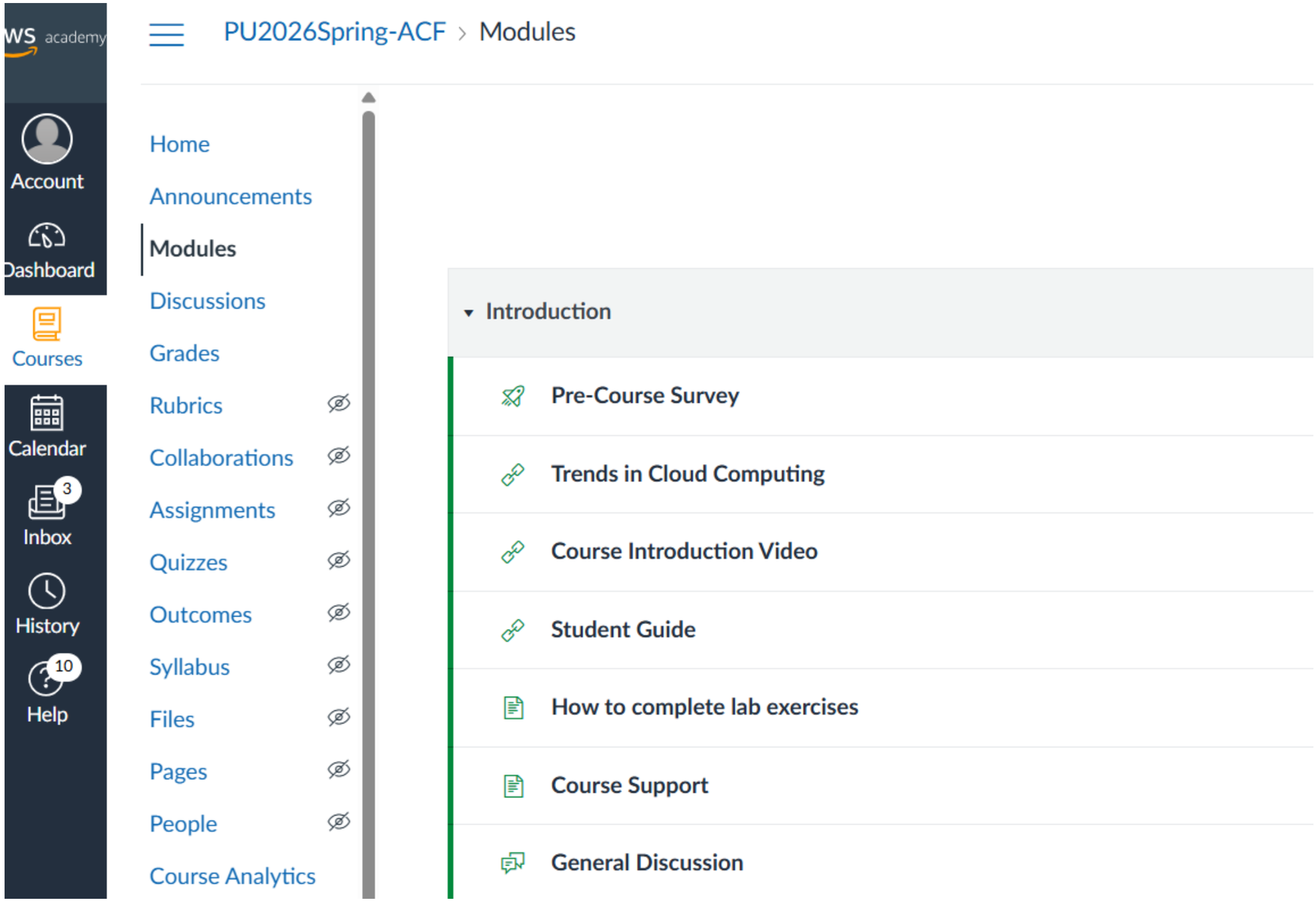
Model	Ownership	Scalability	Security	Cost Model
Public	Provider	★★★★★	★★★	Pay-per-use
Private	Enterprise	★★★	★★★★★	High CapEx
Hybrid	Both	★★★★	★★★★	Balanced
NaaS	Provider	★★★★★	★★★★	Subscription

AWS Academy Courses

Courses

Hidden Courses

MY ACTIVE CLASSES

Course	Nickname	Course Code	Start Date	End Date
AWS Academy Cloud Architecting [160615]	PU2026Spring-ACA	ACAv3EN-US-LTI13-160615	02/22/2026	07/21/2026
AWS Academy Cloud Developing [160614]	PU2026Spring-ACD	ACDv2EN-LTI13-160614	02/22/2026	07/21/2026
AWS Academy Cloud Foundations [160613]	PU2026Spring-ACF	ACFv2EN-LTI13-160613	02/22/2026	07/21/2026

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